

# Morrow Technical Specification

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Status: Experimental

## 1. System Objective

The objective of Morrow is to create a local-first system capable of converting a collection of documents into searchable knowledge.

The system should allow a user to ask natural-language questions and retrieve document sections that are semantically related to the question.

## 2. Input Formats

The initial implementation supports four document types:

PDF

DOCX

TXT

Markdown

Additional formats may be introduced in future milestones.

## 3. Extraction Layer

The extraction layer is responsible for converting different document formats into a common text representation.

PDF extraction uses PyMuPDF.

DOCX extraction reads paragraphs from Microsoft Word documents.

TXT and Markdown files are processed as text.

The extraction layer should hide format-specific implementation details from the rest of the system.

#### 4. Chunking Layer

The chunking layer receives extracted text and divides it into smaller sections.

The current implementation uses word-based chunking.

Experimental parameters include:

Chunk size: 500 words

Overlap: 100 words

The chunking system also validates its parameters.

Chunk size must be greater than zero.

Overlap cannot be negative.

Overlap must be smaller than the chunk size.

## 5. Chunk Metadata

Each chunk contains information that identifies its origin.

The metadata currently includes:

Chunk ID

Chunk index

Source file

File type

Page number

Word count

This information will later help the retrieval system display useful source references.

## 6. Embedding Layer

The embedding layer will convert each chunk into a numerical vector.

The same embedding model must be used consistently when comparing document chunks with user queries.

The project will experiment with different embedding models and vector dimensions.

## 7. Database Layer

PostgreSQL will be used as the primary database.

The pgvector extension will provide vector storage and similarity search functionality.

The database design is expected to separate documents, chunks, embeddings, and metadata where appropriate.

## 8. Retrieval Layer

The retrieval layer will accept a natural-language query.

The query will first be converted into an embedding.

The query embedding will then be compared against stored chunk embeddings.

The system will return the chunks with the highest similarity scores.

## 9. Evaluation

Retrieval quality should not be judged only by whether the system returns an answer.

Future evaluation should measure:

Retrieval precision

Retrieval recall

Context relevance

Answer correctness

Citation accuracy

Search latency

## 10. Long-Term Architecture

The long-term architecture may include:

Document ingestion

Text extraction

Chunking

Embedding generation

Vector storage

Semantic retrieval

Reranking

Context construction

Local language model

Answer generation

Source citation

Morrow is intended to remain modular so individual components can be replaced without redesigning the entire system.